

ENTITIES, ATTRIBUTES, PRIMARY KEYS AND RELATIONSHIPS

For the **E-Commerce Order Management System**, the database must store information about customers, sellers, products, orders, payments, and delivery. Each major object is represented as an **entity**, its properties are represented as **attributes**, and a **primary key** uniquely identifies each record.

1. Customer Entity

The **Customer** entity stores information about users who purchase products from the e-commerce system.

Attributes:

- **Customer_ID** – Unique identification number of the customer.
- **Customer_Name** – Name of the customer.
- **Email** – Customer's email address.
- **Phone** – Contact number.
- **Address** – Customer's address.
- **Registration_Date** – Date on which the customer registered.

Primary Key: Customer_ID

A customer can place multiple orders, but each order belongs to one customer.

2. Seller Entity

The **Seller** entity stores information about sellers who provide products through the platform.

Attributes:

- **Seller_ID** – Unique identification number of the seller.
- **Seller_Name** – Name of the seller.
- **Email** – Seller's email address.
- **Phone** – Contact number.
- **Address** – Seller's address.

Primary Key: Seller_ID

One seller can sell multiple products.

3. Category Entity

The **Category** entity is used to classify products into different groups.

Attributes:

- **Category_ID** – Unique identifier for the category.
- **Category_Name** – Name of the category, such as Electronics, Clothing, or Books.

Primary Key: Category_ID

One category can contain many products.

4. Product Entity

The **Product** entity stores information about products available for purchase.

Attributes:

- **Product_ID** – Unique identification number of the product.
- **Product_Name** – Name of the product.
- **Category_ID** – Identifies the product category.
- **Seller_ID** – Identifies the seller.
- **Price** – Selling price of the product.
- **Stock_Quantity** – Available quantity.
- **Product_Status** – Indicates whether the product is available or unavailable.

Primary Key: Product_ID

Foreign Keys: Category_ID, Seller_ID

Each product belongs to one category and is associated with a seller. A seller can have multiple products.

5. Order Entity

The **Order** entity stores the main information about a customer's purchase.

Attributes:

- **Order_ID** – Unique identification number of the order.

- **Customer_ID** – Identifies the customer who placed the order.
- **Order_Date** – Date on which the order was placed.
- **Order_Status** – Current status of the order.
- **Total_Amount** – Total value of the order.

Primary Key: Order_ID

Foreign Key: Customer_ID

One customer can place many orders. Each order must belong to a valid customer.

6. Order_Details Entity

The **Order_Details** entity stores the individual products included in an order.

Attributes:

- **Order_Detail_ID** – Unique identifier for each order-detail record.
- **Order_ID** – Identifies the related order.
- **Product_ID** – Identifies the ordered product.
- **Quantity** – Number of units ordered.
- **Unit_Price** – Price of one unit.
- **Subtotal** – Total price for that product.

Primary Key: Order_Detail_ID

Foreign Keys: Order_ID, Product_ID

This entity is important because a single order can contain multiple products. It also connects the **Order** and **Product** entities.

Formula:

Subtotal = Quantity × Unit Price

7. Payment Entity

The **Payment** entity stores payment information associated with an order.

Attributes:

- **Payment_ID** – Unique payment identifier.

- **Order_ID** – Identifies the related order.
- **Payment_Method** – Method used for payment.
- **Amount** – Amount paid.
- **Payment_Date** – Date of payment.
- **Payment_Status** – Status such as Pending, Successful, Failed, or Refunded.

Primary Key: Payment_ID

Foreign Key: Order_ID

Payment information must always be associated with a valid order.

8. Shipping Entity

The **Shipping** entity maintains information about the delivery of an order.

Attributes:

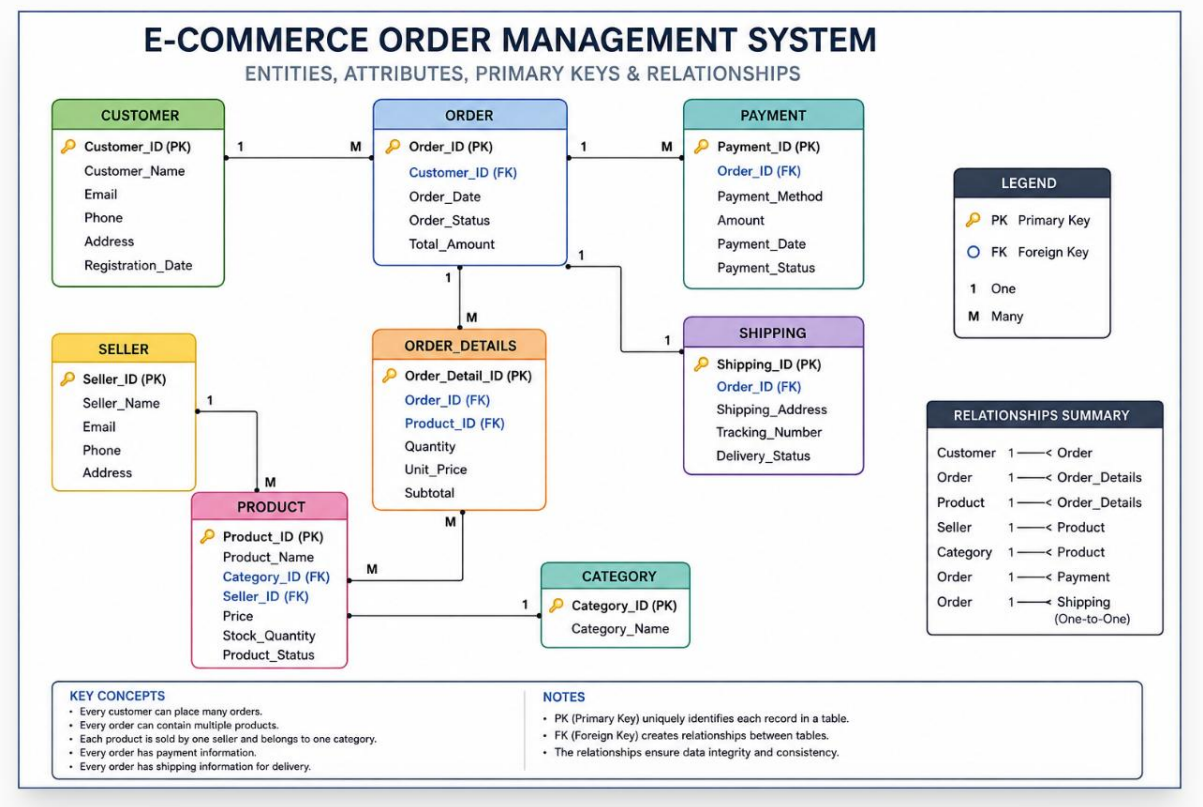
- **Shipping_ID** – Unique shipping identifier.
- **Order_ID** – Identifies the related order.
- **Shipping_Address** – Address where the order must be delivered.
- **Tracking_Number** – Unique tracking information.
- **Delivery_Status** – Current delivery status.

Primary Key: Shipping_ID

Foreign Key: Order_ID

An order can have one shipping record, which is used to track the delivery.

Diagram



Relationships Between Entities

Relationship	Type	Explanation
Customer → Order	1 : M	One customer can place many orders.
Seller → Product	1 : M	One seller can sell many products.
Category → Product	1 : M	One category can contain many products.
Order → Order_Details	1 : M	One order can contain multiple products through order details.
Product → Order_Details	1 : M	One product can appear in many different orders.
Order → Payment	1 : 1 / 1 : M	An order can have associated payment information.
Order → Shipping	1 : 1	An order can have one shipping record.

These relationships form the basic structure of the database and ensure that customer, product, order, payment, and shipping information remains properly connected.

Simple Relationship Flow

Customer

↓ *places*

Order

↓ *contains*

Order_Details

↓ *refers to*

Product

↑

belongs to

Category

Seller → *sells* → **Product**

Order → *has* → **Payment**

Order → *has* → **Shipping**